SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – MCQ Test

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO3 – Precision	**Assessment:**	Assessment Tool 3 – MCQ Test
Weightage:	10% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
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Section / Batch:		**Date:**	

Code of Conduct

I SYED MUHAFIZ A (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Instructions

- Read each question carefully before selecting your answer.
- This assessment consists of 50 multiple-choice questions.
- Do not use external resources, AI tools, textbooks, or assistance from others during the assessment unless explicitly permitted by the instructor.
- Upload Score Card in GCR and submit score card printout.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Context-Free Grammars (CFG)	CFG components, productions, terminals, non-terminals, recursion, sentence generation	1 – 10
Sentence Construction and Agreement	Parse trees, sentence structure, grammatical agreement, syntactic validation	11 – 20
Subcategorization	Verb frames, complement requirements, argument structure analysis	21 – 25
Top-Down Parsing	Parsing strategy, derivation process, recursion issues, grammar expansion	26 – 30
Earley Parsing	Predictor, Scanner, Completer operations, ambiguity handling	31 – 35
Feature Structures	Feature representation, unification, constraint checking, agreement enforcement	36 – 40
Probabilistic Context-Free Grammars (PCFG)	Rule probabilities, ambiguity resolution, probabilistic parsing	41 – 48
Integrated Syntax Analysis	CFG + Feature Structures + PCFG integration in NLP applications	49 – 50

Annexure A – MCQ Test

1. What is the primary goal of Natural Language Processing (NLP)?
A) To increase computer hardware speed
B) To enable computers to process and understand human language
C) To design operating systems
D) To manage computer networks

2. Which model is commonly used to describe whether a string belongs to a particular language?
A) Database model
B) Mathematical model
C) Financial model
D) Hardware model

3. Which symbol is commonly used in regular expressions to represent zero or more occurrences of the preceding element?
A) +
B) ?
C) *
D) ^

4. Which regular expression represents a sequence containing one or more occurrences of a?
A) a*
B) a+
C) a?
D) a|

5. What is the main function of a Finite State Automaton (FSA)?
A) To store large databases
B) To recognize patterns in strings
C) To translate programming languages
D) To generate images

6. Which type of morphology deals with grammatical variations such as tense and number?
A) Derivational morphology

B) Inflectional morphology

C) Computational morphology

D) Structural morphology

7. Which example represents derivational morphology?

A) cat → cats

B) walk → walked

C) happy → happiness

D) book → books

8. What is the purpose of finite-state morphological parsing?

A) To identify the internal morphological structure of words

B) To identify network protocols

C) To calculate sentence length

D) To generate numerical data

9. What does the Porter Stemmer primarily perform?

A) Sentence parsing

B) Word stemming

C) POS tagging

D) Speech recognition

10. The word “unhappiness” can be analyzed as un + happy + ness. Which process is illustrated by this analysis?

A) Inflectional morphology only

B) Derivational morphology

C) Syntax parsing

D) Regular expression matching

11. What does an N-gram model estimate?

A) Probability of a word based on previous words

B) Number of letters in a word

C) Meaning of a paragraph

D) Number of sentences in a document

12. A bigram model considers how many words at a time?

A) One

B) Two

C) Three

D) Four

13. What is the main problem with an unsmoothed N-gram model?

A) It cannot count words

B) It assigns zero probability to unseen N-grams

C) It cannot recognize nouns

D) It cannot process sentences

14. What is the primary purpose of smoothing in language models?

A) To remove punctuation

B) To assign probabilities to unseen events

C) To identify verbs

D) To shorten sentences

15. What does entropy measure in an N-gram language model?

A) Linguistic uncertainty

B) Number of nouns

C) Sentence length

D) Number of characters

16. Which word class does “quickly” belong to in the sentence “She runs quickly”?

A) Noun

B) Verb

C) Adjective

D) Adverb

17. What is the purpose of a tagset in POS tagging?

A) To define the set of grammatical labels used for words

B) To store sentences

C) To generate regular expressions

D) To calculate entropy

18. In the sentence “Book a ticket,” what is the POS of “Book”?

A) Noun

B) Verb

C) Adjective

D) Adverb

19. Which POS tagging approach uses manually written linguistic rules?

A) Stochastic tagging

B) Rule-based tagging

C) Transformation-based tagging

D) Random tagging

20. Which POS tagging method combines an initial tagging with a sequence of corrective transformations?

A) Rule-based tagging

B) Stochastic tagging

C) Transformation-based tagging

D) Dictionary-based tagging

21. What is the primary purpose of a Context-Free Grammar (CFG) in NLP?

A) To represent syntactic structure of sentences

B) To calculate word frequency

C) To identify word meanings

D) To translate documents

22. Which component of a CFG specifies how a non-terminal can be expanded?

A) Terminal symbol

B) Production rule

C) Probability table

D) Lexicon only

23. In the rule S → NP VP, what does S usually represent?

A) Sentence

B) Subject

C) Semantics

D) String

24. Which structure visually represents the syntactic organization of a sentence?

A) Bar chart

B) Parse tree

C) Flowchart

D) Frequency table

25. What does grammatical agreement ensure?

A) Words have the same number of letters

B) Related words have compatible grammatical features

C) All words have the same meaning

D) Sentences contain equal numbers of words

26. What does subcategorization describe?

A) The arguments or complements that a word, especially a verb, requires

B) The number of letters in a word

C) The probability of a sentence

D) The meaning of punctuation

27. What is parsing in NLP?

A) Converting text into numbers only

B) Determining the syntactic structure of a sentence

C) Removing stop words

D) Translating words between languages

28. Which parsing strategy begins with the start symbol and attempts to generate the input sentence?

A) Bottom-up parsing

B) Top-down parsing

C) Stochastic tagging

D) Semantic parsing

29. Which parsing algorithm can parse sentences using dynamic programming and handle left-recursive grammars?

A) Porter algorithm

B) Earley parser

C) Viterbi tagger

D) PageRank algorithm

30. What is the purpose of a Probabilistic Context-Free Grammar (PCFG)?

A) To associate probabilities with grammar productions

B) To remove grammar rules

C) To identify word stems

D) To classify documents

31. What is the main objective of semantic analysis?

A) To determine the meaning of linguistic expressions

B) To count characters

C) To identify punctuation

D) To format documents

32. Which formal system is commonly used to represent relationships between entities and properties?

A) First-Order Predicate Calculus

B) Regular Expression

C) Finite Automaton

D) N-gram model

33. In first-order predicate logic, what does P(x) typically represent?

A) A probability

B) A predicate applied to an argument

C) A parsing rule

D) A POS tag

34. Which representation can express the statement “John loves Mary” as a relationship between entities?

A) loves(John, Mary)

B) John + Mary

C) John/Mary

D) John = Mary

35. What is syntax-driven semantic analysis?

A) Deriving meaning representations based on syntactic structure

B) Counting syntactic words

C) Removing grammatical rules

D) Generating random meanings

36. What is a lexeme?

A) An abstract vocabulary unit representing related word forms

B) A punctuation mark

C) A sentence structure

D) A grammar rule

37. What does word sense refer to?

A) The grammatical category of a word

B) A particular meaning of a word

C) The number of letters in a word

D) The pronunciation of a word

38. What is the main goal of Word Sense Disambiguation (WSD)?

A) To determine the intended meaning of an ambiguous word

B) To identify sentence boundaries

C) To count words

D) To remove prefixes

39. In “The fisherman sat on the bank,” which meaning of “bank” is most appropriate?

A) Financial institution

B) River side

C) Storage facility

D) Computer memory

40. What is the primary goal of Information Retrieval (IR)?

A) To retrieve relevant information from a collection of documents

B) To generate grammar rules

C) To identify speech sounds

D) To stem every word

41. What does discourse analysis primarily study?

A) Language beyond individual sentences and its context

B) Individual characters only

C) Computer hardware

D) Word spelling only

42. What is reference resolution?

A) Determining what an expression such as “he” or “it” refers to

B) Finding the number of words in a sentence

C) Identifying punctuation marks

D) Translating a sentence

43. In the sentence “Ravi bought a car. He likes it,” what does “He” refer to?

A) Car

B) Ravi

C) Likes

D) It

44. What does text coherence refer to?

A) The meaningful and logical relationship among parts of a text

B) The number of words in a paragraph

C) The spelling of words

D) The number of punctuation marks

45. What is a discourse act or dialogue act?

A) The communicative function performed by an utterance

B) The number of words in dialogue

C) A grammatical error

D) A punctuation rule

46. Which of the following is an example of a dialogue act?

A) Request

B) Noun

C) Prefix

D) Stem

47. What is the main purpose of a conversational agent?

A) To interact with users through natural language

B) To perform only mathematical calculations

C) To store images

D) To compile source code

48. What is the purpose of surface realization in language generation?

A) To convert an abstract representation into natural-language text

B) To identify word senses

C) To parse a sentence

D) To calculate N-gram entropy

49. Which machine translation approach uses an intermediate language-independent representation?

A) Direct translation

B) Interlingua

C) Word-for-word translation

D) Rule deletion

50. What do statistical approaches to machine translation primarily learn from?

A) Parallel and bilingual language data

B) Only punctuation marks

C) Computer hardware specifications

D) Regular expressions alone

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Number of Correct Answers	100%	45 to 50 correct answers	38 to 44 correct answers	25 to 37 correct answers	0 to 24 correct answers
Accuracy	100%	Excellent (90–100%)	Good (75–89%)	Average (50–74%)	Poor (Below 50%)

No. of Questions	Number of Correct Answers	Accuracy	Total %
50			

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________